Evolutionary Aircraft Design Optimisation - Report

3.1 Problem Understanding

The task is to find aircraft configurations that simultaneously minimise aerodynamic drag, maximise fuel volume stored in the wings, and maximise cargo space in the fuselage. These objectives conflict: thinner wings reduce drag but hold less fuel, a wider fuselage increases cargo but also zero-lift drag, a larger wing area helps lift and fuel volume but increases drag. Because of these conflicts, no single design is best in all three objectives. The answer is a set of trade-off designs, the Pareto front, where improving one objective necessarily worsens another.

A multi-objective evolutionary algorithm is the right tool here because it searches for all trade-offs at once without requiring weights to combine the objectives. A single-objective optimiser would need those weights in advance, which is a subjective choice the problem statement deliberately avoids.

3.2 Design of the Algorithm

Algorithm: NSGA-II. It was designed for continuous multi-objective problems exactly like this one. It finds diverse Pareto-optimal solutions through two mechanisms: Pareto ranking (measures solution quality) and crowding distance (measures how isolated a solution is, preserving spread along the front).

Encoding: Each aircraft is a real-valued vector of 6 variables: wing span b [20-80 m], wing area Sw [80-300 m²], sweep angle Λ [0-45°], thickness-to-chord ratio tc [0.08–0.20], fuselage length Lf [20-60 m], fuselage diameter Df [3-7 m]. AR is computed as b^2/Sw rather than stored as a free variable.

Initialisation: 100 individuals sampled uniformly at random within the variable bounds.

Objectives: evaluate_aircraft(x) computes C_D, V_fuel, V_cargo from the notebook's physics models. The objective vector is [C_D, -V_fuel, -V_cargo] - negatives convert maximisation to minimisation.

Constraints: Handled via a penalty function. constraint_violation(x) squares each violation (min fuel, min cargo, AR bounds, Sw bounds) and sums them. During evolution: f_penalised = f + 10000·P(x). At the end, only designs with zero violation are kept for the final Pareto front.

Selection: Binary tournament - two random candidates compared by Pareto rank, ties broken by crowding distance. This simultaneously favours quality and diversity.

Crossover: SBX (η_c = 15, prob = 0.9) - produces children near the parents in real-valued space, each gene crossed independently with probability 0.5.

Mutation: Polynomial mutation (η_m = 20, rate = 1/6) - small random perturbation to each gene, clipped to bounds.

Diversity: Crowding distance. When a front partially fits the next generation, the most isolated solutions are selected first, keeping the front spread out.

Final selection: After 200 generations, feasible designs are extracted, non-dominated sorted, and front 0 is the output.

3.3 Parameter Choices

Parameter	Value	Justification
Population size	100	Enough diversity for 6 variables; larger populations improve front quality but increase runtime.
Generations	200	Convergence observed before gen 200 in all runs; stopping earlier risks missing good solutions.
Crossover prob.	0.9	High recombination rate, standard for SBX on continuous problems.
Mutation rate	1/n	Each gene mutated with prob. 1/6. Standard recommendation for polynomial mutation.
SBX index (ηc)	15	Moderate spread - offspring close to but distinct from parents.
Mutation index (ηm)	20	Small typical steps; fine-tunes solutions without destroying good genes.
Penalty weight (λ)	10 000	Ensures any infeasible design ranks below any feasible one on every objective.

High crossover probability drives recombination of existing solutions. Mutation is the main source of new variation - too high destroys good solutions; too low causes early stagnation. The penalty weight was verified by checking that all final Pareto-front designs have zero constraint violation.

3.4 Results

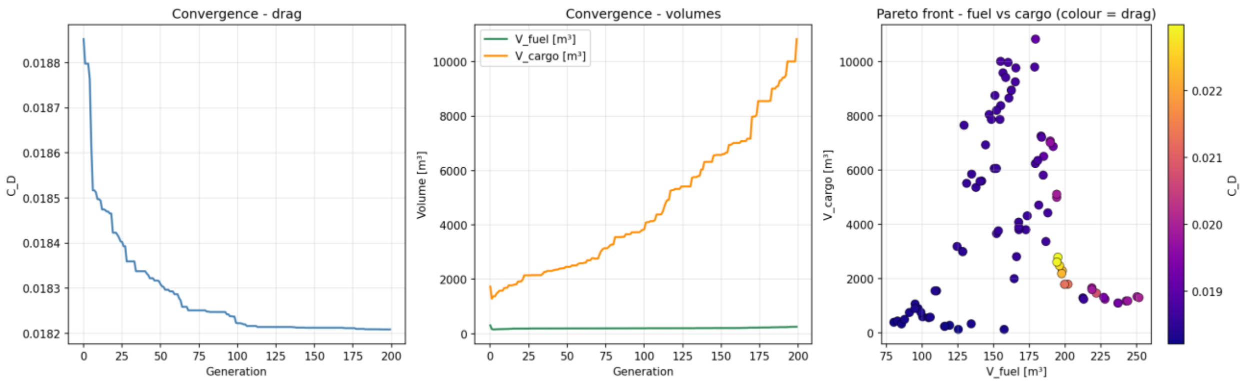

Figure 1. Left: drag convergence. Centre: volume convergence. Right: Pareto front (colour = drag).

Drag converges within ~50 generations and stabilises. Fuel and cargo volumes improve steadily throughout. The algorithm found 98 feasible, non-dominated designs. The Pareto front spans cargo volumes from ~120 m³ to ~2800 m³ and fuel volumes from 85 m³ to 260 m³, with drag tightly clustered between 0.0182 and 0.0196.

b [m]	Sw [m²]	tc	Lf [m]	Df [m]	AR	CD	Vfuel [m³]	Vcargo [m³]
64.8	300	0.183	60.4	1.83	13.99	0.01821	114.5	121.6
64.3	299.5	0.135	60	5.4	13.8	0.01833	85	1049
45.6	300	0.259	60	7.15	6.94	0.01854	229.6	1839.3
56.9	292	0.201	60	7	11.1	0.01843	135.3	1764
42.9	300	0.276	60	6.32	6.14	0.01856	260.1	1438.3
42.9	300	0.235	60	8.83	6.13	0.01956	222.1	2805.4

Table 1. Representative Pareto-optimal designs - all feasible and non-dominated.

Row 1 is the minimum-drag design: high AR (14), large wing, thin fuselage - barely meets the cargo minimum. Rows 5–6 are the high-cargo end: fuselage diameter ~8.8 m drives cargo above 2800 m^3 at a small drag cost.

3.5 Discussion

Influential variables: Wing area Sw is at its upper bound (300 m^2) in almost every design - the algorithm pushes it to the maximum because it simultaneously reduces induced drag (higher AR) and increases fuel volume. Fuselage diameter Df is the dominant driver of cargo volume and the main source of variation across the front. Thickness-to-chord ratio tc controls fuel volume with relatively little drag penalty.

Trade-offs: The clearest tension is drag vs cargo - a slender fuselage minimises drag but limits cargo. Fuel volume is more independent; it can be increased by thickening the wings without heavily penalising either other objective.

Feasibility and diversity: All 98 final designs are fully feasible. The front is diverse in cargo and fuel volume; drag variation is small because feasibility constraints tightly bound the achievable range.

Limitations: The model has no structural weight feedback (heavier fuselages would increase required lift and drag), no compressibility effects, and the stability and stress constraints are not numerically implemented. The large fuselage diameters on some high-cargo designs (8–9 m) would likely be unrealistic in practice.

Possible improvements: Add a structural weight model; implement the stability constraint; increase population to 200 and generations to 500; try NSGA-III, which handles three-objective problems with better front coverage.

3.6 Defense of Design Choices

- **Why NSGA-II?** Three continuous, conflicting objectives, no single correct answer - exactly the class of problem NSGA-II was designed for. It requires no objective weights and produces a diverse front in one run.

- **Why these parameters?** They follow the recommendations in the original NSGA-II paper and have been validated on engineering design problems of similar dimensionality.
- **How were constraints handled?** Penalty function with $\lambda = 10\,000$, verified effective - all 98 final designs have zero violation.

- **Are the results meaningful?** Yes - convergence plots show stabilisation well before the stopping criterion, and the results match physical intuition (high AR -> low drag, wide fuselage -> high cargo).

- **Effect of higher mutation:** More exploration but slower convergence - good solutions are disrupted before being exploited.

- **Effect of lower mutation:** Risk of premature convergence - the population clusters and misses parts of the Pareto front.

- **Weaknesses:** Simplified physics, unimplemented constraints, $O(n^2)$ sort cost limits scalability to larger populations.